

Complex Terrain Load Carriage Energy Expenditure Estimation Using Global Positioning System Devices

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ABSTRACT

POTTER, A. W., W. R. SANTEE, S. P. MULLEN, A. J. KARIS, L. A. BLANCHARD, M. N. ROME, K. P. PITTS, and D. P. LOONEY. Complex Terrain Load Carriage Energy Expenditure Estimation Using Global Positioning System Devices. *Med. Sci. Sports Exerc.*, Vol. 50, No. 10, pp. 2145–2149, 2018. **Introduction:** Military load carriage can cause extreme energy expenditure (EE) that is difficult to estimate due to complex terrain grades and surfaces. Global Positioning System (GPS) devices capture rapid changes in walking speed and terrain but the delayed respiratory response to movement is problematic. We investigated the accuracy using GPS data in three different equations to estimate EE during complex terrain load carriage. **Methods:** Twelve active duty military personnel (age, 20 ± 3 yr; height, 174 ± 8 cm; body mass, 85 ± 13 kg) hiked a complex terrain trail on multiple visits under different external load conditions. Energy expenditure was estimated by inputting GPS data into three different equations: the Pandolf–Santee equation, a recent GPS-based equation from de Müllenheim et al.; and the Minimum Mechanics model. Minute-by-minute EE estimates were exponentially smoothed using smoothing factors between 0.05 and 0.95 and compared with mobile metabolic sensor EE measurements. **Results:** The Pandolf–Santee equation had no significant estimation bias (-2 ± 12 W; $P = 0.89$). Significant biases were detected for the de Müllenheim equation (38 ± 13 W; $P = 0.004$) and the Minimum Mechanics model (-101 ± 7 W; $P < 0.001$). **Conclusions:** Energy expenditure can be accurately estimated from GPS data using the Pandolf–Santee equation. Applying a basic exponential smoothing factor of 0.5 to GPS data enables more precise tracking of EE during non–steady-state exercise. **Key Words:** METABOLIC RATE, MILITARY, PERFORMANCE, ENERGY COST, PREDICTIVE EQUATIONS

Military load carriage can cause extreme energy expenditure (EE) (1) and impair Warfighter performance (2–4). Energy expenditure estimation equations were developed from laboratory research to quantify load carriage economy (5–9), yet the steady-state laboratory walking protocols used differ from the load carriage performed in military field operations. Warfighters often carry heavy loads over complex terrain with variable grades and surfaces (10,11). In addition to direct effects on EE, complex terrain

features have indirect effects due to their influence on walking pace.

Complex terrain load carriage must be properly characterized for accurate EE estimates but is difficult to replicate in the laboratory. Estimating EE with trail averages is a simple but imprecise approach. Energy expenditure is disproportionately increased by faster walking speeds (6,9,12). Thus, inputting the average speed of a prolonged walk will underestimate EE if speed varies considerably. Grade’s “J-shaped” effect on EE (7,13) is lost if averaged over an undulating trail. Terrain surfaces exacerbate these issues by multiplying energy costs by up to 3.5 times (14,15). Consequently, proper characterization of complex terrain load carriage requires finer detail and higher-resolution data.

Global positioning system (GPS) devices are a potential solution because they can capture near instantaneous changes in walking speed and terrain. However, the delayed respiratory response to physical activity (16,17) complicates the use of walking equations on a second-by-second basis. Energy expenditure will be overestimated initially before a respiratory steady-state is reached. Excess postexercise oxygen consumption is a well-documented phenomenon (18,19) that will be underestimated by walking equations when movement

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TABLE 1. Energy expenditure estimation equations.

Equation	Output	Formula
Pandolf-Santee (8,9)	W	$1.5 M + 2.0 (M + L) (L/M)^2 + \eta (M + L) (1.5 V^2 + 0.35 VG) - D\eta [(G (M + L) V)/3.5 - ((M + L) (G + 6)^2)/M] + (25 S^2)]$
de Müllenheim (21)	$\text{mL } \dot{V}\text{O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$	$\eta (2.4037 + 2.5107 V + 0.2828 VG)$
Minimum Mechanics (8,9)	$\text{mL } \dot{V}\text{O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$	$G \geq 0: \dot{V}\text{O}_{2\text{-rest}} + \eta (0.32 G + 3.28 + (1 + (0.19 G))(2.66 V^2))$ $G < 0: \dot{V}\text{O}_{2\text{-rest}} + \eta 0.73(3.28 + 2.66 V^2)$

M, body mass (kg); L, external load (kg); η , terrain factor; V, speed ($\text{km} \cdot \text{h}^{-1}$ for de Müllenheim or $\text{m} \cdot \text{s}^{-1}$ otherwise); G, grade (%); D, 1 for downhill or 0 otherwise; $\dot{V}\text{O}_{2\text{-rest}}$, resting metabolic rate calculated from Schofield (22).

ceases. Estimation precision may be improved by applying an exponential smoothing filter to correct for the delayed respiratory response to physical activity changes. Exponential smoothing is a time series technique similar to a moving average that applies an exponentially greater weight to more recent data (20). Applying a recursive exponential filter will introduce a lag that could better align GPS-based EE estimates with observed respiratory values.

We investigated whether the energy expended during complex terrain load carriage could be accurately estimated by inputting GPS data into three equations: the Santee-corrected Pandolf equation (6,7); a recent GPS-based equation from de Müllenheim et al. (21); and the Minimum Mechanics model from Ludlow and Weyand (9). Our study also examined if exponential smoothing the GPS-based estimates would correct for the delayed respiratory response to locomotion.

METHODS

Study design. Active military personnel hiked a complex terrain trail on multiple visits under different external load conditions (0%, 30%, or 45% of the individual's body mass). The 0% body mass load was carried on the first visit due to safety concerns, whereas the 30% and 45% body mass loads were carried in a random order on separate visits. Energy expenditure was measured using a mobile metabolic sensor and estimated using three different equations (Table 1). Body mass and external load were measured daily on a digital scale. Speed and grade ($\Delta\text{altitude}/\text{distance}$) were calculated over 10-s periods. We zoned off each paved and dirt trail segment by GPS coordinates prior to data collection. The terrain factor updated whenever a participant moved into a paved or dirt road zone.

Participants. Twelve (1 woman, 11 men) active duty military personnel (age, 20 ± 3 yr; height, 174 ± 8 cm; body mass, 85 ± 13 kg) volunteered for this study. Written informed consent was obtained from all participants before data collection. All participants were medically cleared before participation, passed their most recent Army Physical Fitness Test, and had no recent history of musculoskeletal injury or other preexisting conditions which would preclude their participation in the study. This study was approved by both the scientific review committee and institutional review board at the United States Army Research Institute of Environmental Medicine (USARIEM; Natick, MA).

Procedures. Nude bodyweights were measured daily on a calibrated digital scale. Participants completed a brief warm-up walk before being instrumented with a mobile metabolic sensor system (COSMED K4_b² (K₄) COSMED Sri., Rome, Italy) consisting of a portable unit, battery, face mask covering the nose and mouth, mounting harness, connecting

tubes, and cables. A 1-Hz GPS monitor (Garmin Foretrex 301; Garmin, Olathe, KS) was placed on each shoulder to collect positional data. Global Positioning System accuracy was considered acceptable throughout data collection as the number of locked satellites remained above 4 and horizontal dilution of precision remained below 3. Participants were then given the test visit's assigned load. The 0% body mass load included the Army Combat Uniform uniform and personal boots. The 30% body mass load added a replica rifle, body armor, water, and simulated munitions. The 45% load added a Modular Lightweight Load Carrying Equipment (MOLLE) frame rucksack. Additional weight was added to the rucksack and/or vest pouches to match load to the individual body mass percentage.

All data were collected on the same 2.5-km complex terrain trail (Natick Soldier System Center Fitness Trail, Natick, MA) (Fig. 1). We placed a physical marker on each terrain transition on the trail and recorded the GPS coordinates before data collection. Paved and dirt trail segments were assigned terrain factors of 1.0 and 1.2, respectively (14). Grades were determined from indexing altitudes from the U.S. Geological Survey's National Elevation Data Set database using GPS latitude and longitude coordinates. Each participant was required to hike 1 trail lap on the first visit and two to four laps on subsequent visits as fast as possible without jogging, running, or shuffling. Every participant hiked odd numbered laps in the clockwise

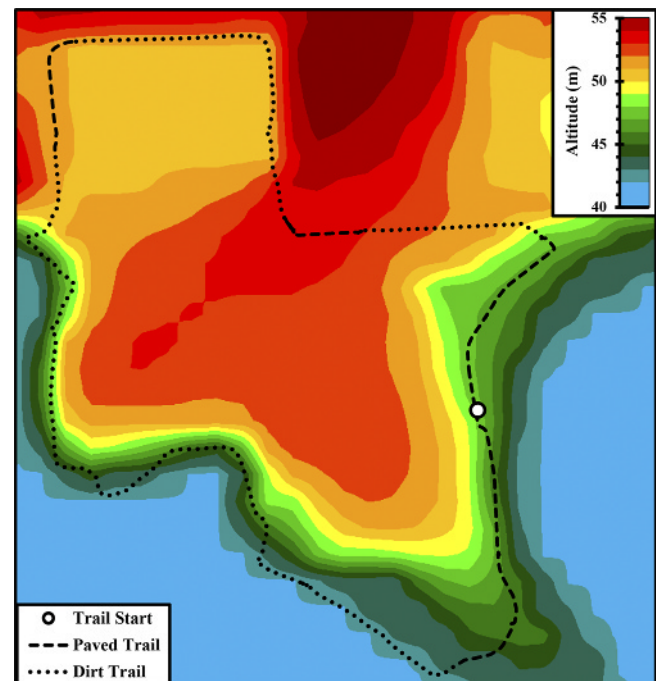


FIGURE 1—Trail map showing topography and surface conditions.

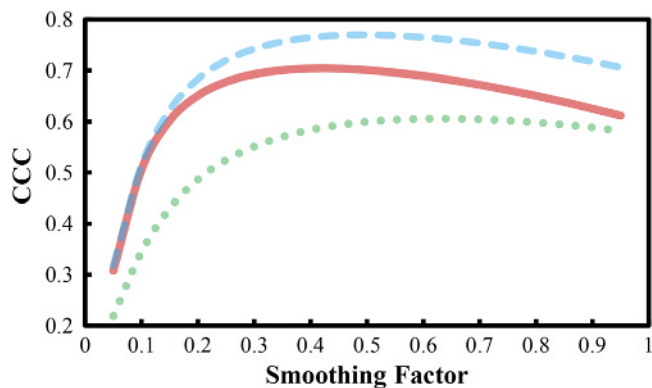


FIGURE 2—CCC across exponential smoothing factors for each equation. Note: dashed blue line, Pandolf-Santee equation; solid red line, de Müllenheim equation; dotted green line, Minimum Mechanics.

direction and even numbered laps in the counterclockwise direction. Participants were permitted to sit down, unmask, and drink water during 5-min rest intervals between laps.

Statistical analyses. All data are displayed as mean $\pm$ SD unless specifically noted otherwise and analyzed using R (Version 3.3.1; R Foundation for Statistical Computing, Vienna, Austria) (23). We fitted four-level mixed effects regression models with random intercepts to produce a properly weighted bias estimate for each predictive equation. Minute-by-minute paired differences (level 1) were analyzed with random effects of trail lap (level 2) within load condition (level 3) within participant (level 4) to account for the dependence among repeated measures as well as unequal data points between laps, loads, and participants. Mixed effects models were analyzed with the “lme4” package (24) and fitted by restricted maximal likelihood. Assumptions of constant variance and normality of errors were checked and validated. Estimated oxygen consumptions were converted to watts with the equation: ($5\dot{V}O_2 \cdot 0.0143^{-1}$). Energy expenditure measurements by the K_4 breath-by-breath and estimates from second-by-second GPS data were both averaged per minute. Minute-by-minute EE estimates were then exponentially smoothed using smoothing

factor weights from 0.05 to 0.95. The smoothed EE estimate at each time point (s_t) was calculated using the following equation:

$$s_t = a u_t + (1 - a) s_{t-1}$$

where a is the smoothing factor weight; u_t , unsmoothed EE estimate.

Concordance correlation coefficients (CCC) were calculated (25) for each smoothing factor weight using the “cccrm” package (26) to determine the best smoothing factor for each equation. The CCC evaluates the accuracy and precision between two measures (27) and is similar to intraclass correlation coefficients used for reliability analysis. Estimation accuracy and precision were evaluated by the bias and residual standard deviation. Modified Bland-Altman plots were analyzed (27,28) with bias estimates and 95% limits of agreement calculated from the linear mixed effects models. This approach accounts for repeated measures as well as participant-, load-, and lap-specific effects on bias estimates. Statistical equivalence between measured and estimated EE was evaluated using the two 1-sided t test (28) with the acceptance criterion limit for statistical equivalence set to 50 W. The level of statistical significance was set as $P \leq 0.05$.

RESULTS

Concordance correlation coefficients for each equation across various exponential smoothing factors are shown in Figure 2. Observed EE was correlated with unsmoothed estimates from the Pandolf-Santee equation (CCC, 0.69), de Müllenheim equation (CCC, 0.60), and Minimum Mechanics (CCC, 0.58). However, the level of agreement was maximized with a smoothing factor of 0.5 for the Pandolf-Santee equation (CCC, 0.77) and de Müllenheim equation (CCC, 0.70) and a smoothing factor of 0.6 for the Minimum Mechanics model (CCC, 0.61).

Figure 3 shows Bland-Altman plots of agreement between measured EE and each equation’s estimates. Pandolf-Santee equation estimates had no significant bias and were

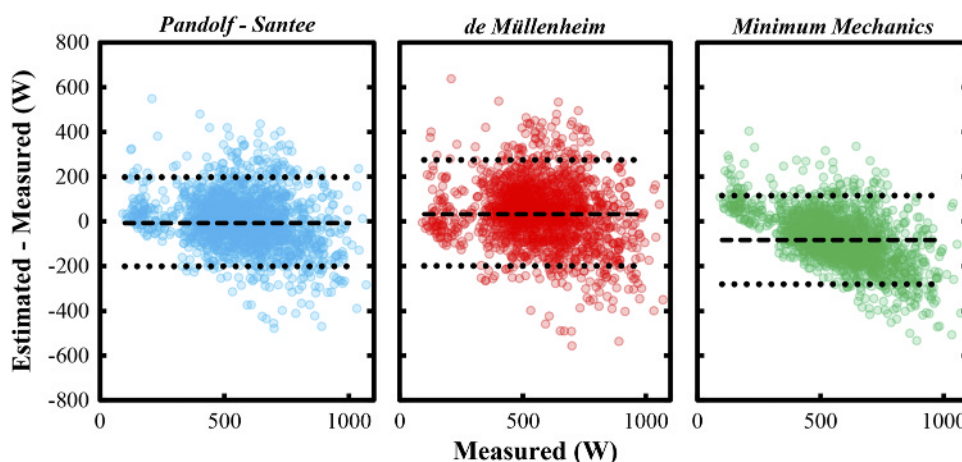


FIGURE 3—Bland-Altman plots of agreement between measured energy expenditure and each equation’s estimates. Note: solid black line, zero line; dotted lines, 95% limits of agreement between estimated and measured energy expenditure.

most precise (bias $\pm$ SD, -7 ± 101 W; $P = 0.635$) and statistically equivalent to measurements ($P < 0.001$). Minimum Mechanics estimates had a significant negative bias (bias $\pm$ SD, -107 ± 102 W; $P < 0.001$) and were not statistically equivalent to measurements ($P > 0.99$). Conversely, the de Müllenheim equation had a significant positive bias and was the least precise (bias $\pm$ SD, 33 ± 121 W; $P = 0.039$) but were statistically equivalent to measurements ($P = 0.003$).

DISCUSSION

The Pandolf–Santee equation accurately estimates EE from GPS data during complex terrain load carriage. The de Müllenheim equation and Minimum Mechanics model overestimated and underestimated EE during the complex terrain course respectively. Our findings show that exponential smoothing improves precision by correcting for the delayed respiratory response to physical activity. Estimation accuracy and precision for these equations were comparable to various physical activity monitors used to estimate whole-room indirect calorimeter EE (29–33) which was unexpected given the added experimental control of these laboratory studies.

The Pandolf equation has been shown to be a relatively accurate predictor of metabolic costs (34) but has also underestimated certain types of military load carriage (12,35). Additionally, these particular studies have shown including external load will not account for metabolic effects of non-typical clothing (e.g., personal protective clothing, explosive ordnance disposal suits). General walking equations are expected to poorly predict EE in these instances without adjustments (36).

The de Müllenheim equation may have overestimated EE because it does not contain a true resting EE component. In the original study (21), all $\dot{V}O_2$ data were collected during walking and divided by both terrain factor and total mass before predictive equations were regressed on it. This means that the equation's intercept must be multiplied by terrain factor and total mass to determine total $\dot{V}O_2$. However, resting EE is the energy required to maintain homeostasis during inactivity and does not change with terrain factor or external load. Furthermore, the original study did not include any resting EE data and had lower terrain factors ($\eta \leq 1.12$) as well as external loads.

The Minimum Mechanics model likely underestimated EE because it simplifies all downhill walking metabolic costs to 73% of level walking EE. Energy expenditure gradually decreases with negative grades to its nadir approximately -8% grade (37) before increasing past level walking at more negative grades (13). Conversely, the Minimum Mechanics model predicts a drastic 27% EE reduction from level walking to a -1% grade. We noted that Minimum Mechanics model accuracy and precision improved when we substituted 0% grade in for all negative grades encountered over the course (bias $\pm$ SD, -54 ± 95 W). A downhill correction factor is necessary to properly estimate EE at slight declines.

This study used a basic exponential smoothing function to improve EE estimation precision. Although this proved to be an elegant correction to the delayed respiratory response to

locomotion, more elaborate functions may further improve EE estimation. For instance, recent statistical methods have shown promise for estimating EE *post hoc* (38). Other body-worn sensors have been used to estimate EE via noninvasive measures, such as foot ground contact (39), heart rate, and accelerometers (40). In field conditions, there are ultimately tradeoffs between the more accurate methods (e.g., more sensors, more invasive or direct measures) and the more practical or simplistic (e.g., less sensors, analytical approaches).

The optimal exponential smoothing factor (0.5) will need to be adjusted when estimating EE over different time epochs. Nevertheless, this study demonstrated considerable improvements in estimation precision during outdoor complex terrain navigation. Intense intermittent exercises, such as competitive soccer (41), would be a challenging but highly impactful area for future research to investigate using exponential smoothing to track metabolic demands.

Rest interval EE was not monitored in this study since participants unmasked to drink water. There may have been larger differences in accuracy and precision between the unsmoothed and smoothed estimates if resting data were collected. On the other hand, the vast majority of $\dot{V}O_2$ recovery occurs within the first few minutes postexercise (19). Consequently, estimation errors would be pronounced but only for the first few minutes of the rest interval.

Global Positioning System is not a tactically acceptable tool to be used during operational missions in its current commercial form. There are detectability, hack-ability, or general security concerns from an operational perspective. However, GPS is currently still widely used operationally, during training, and for personal use. We acknowledge there is a larger issue at play with respect to military use of GPS for certain scenarios. However, the methods from this article could be used currently during training environments or potentially translated for use in future GPS-like systems made for the operational environments.

CONCLUSIONS

Energy expenditure can be accurately estimated from GPS data using the Pandolf–Santee equation. The accuracy of the de Müllenheim equation and Minimum Mechanics model can be improved with corrective factors for downhill grades. Applying a basic exponential smoothing factor of 0.5 to minute-by-minute GPS data enables more precise tracking of EE during non-steady-state exercise.

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The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation, and the results of the present study do not constitute endorsement by ACSM.

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